Machine learning as a tool to map the world?
Challenges and perspectives of spatial predictive mapping of the environment

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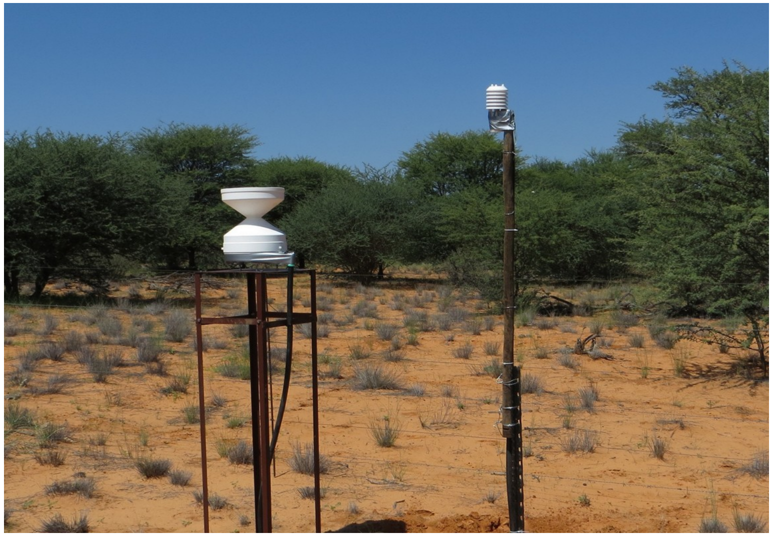
Increasing relevance of environmental monitoring

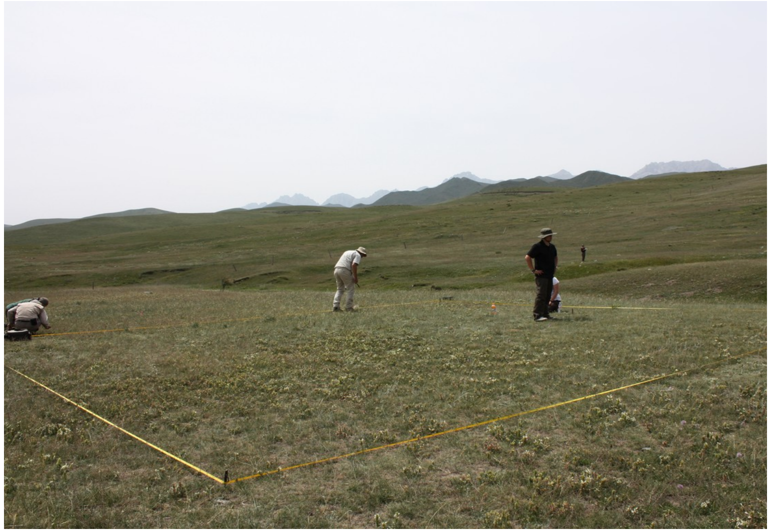
Stefan Doerr (distributed via imaggeo.egu.eu)

Von Aconcagua (talk) - Eigenes Werk, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=6358217

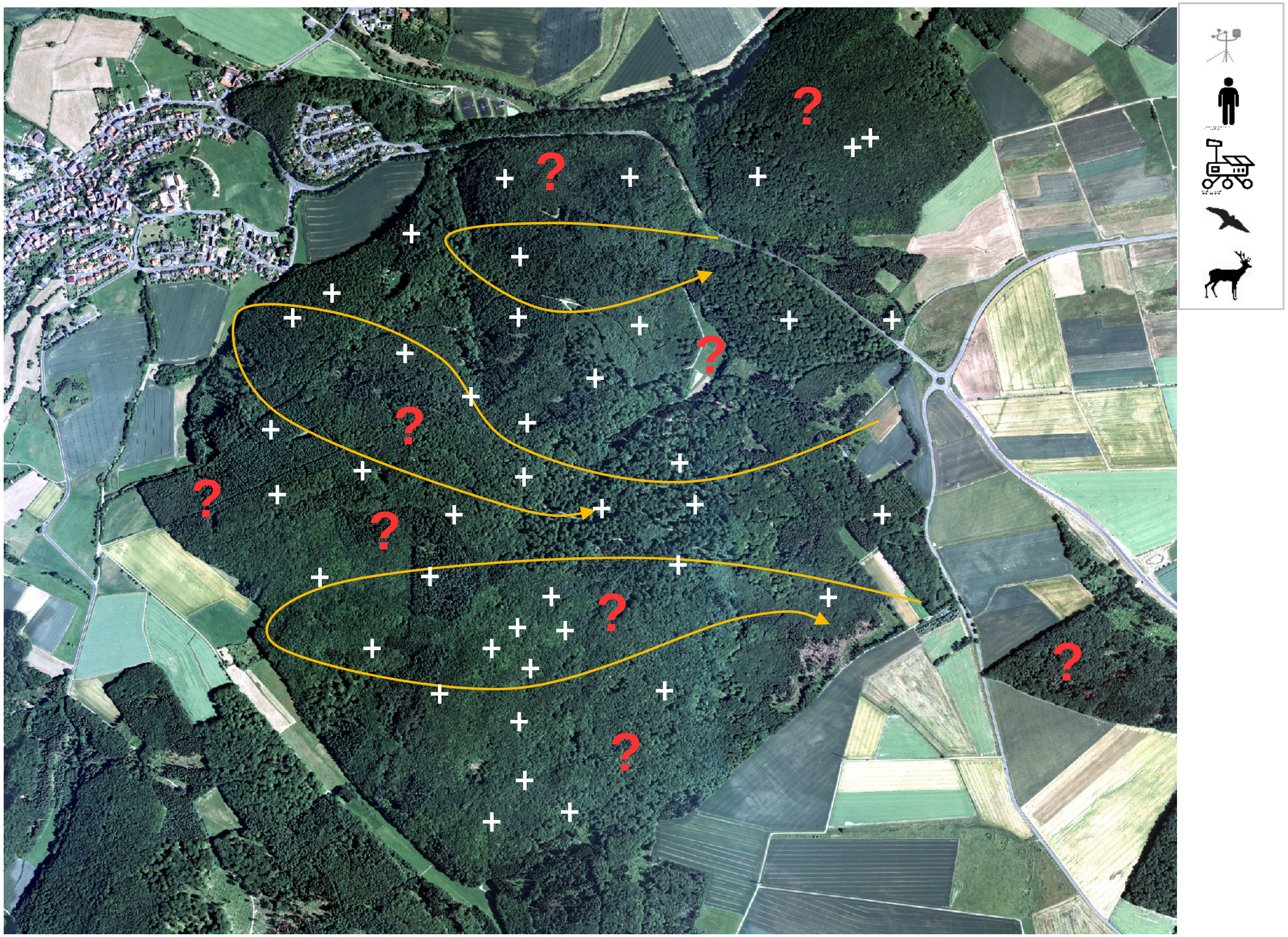
We need spatio-temporal information about the environment!

...how do we monitor the environment?

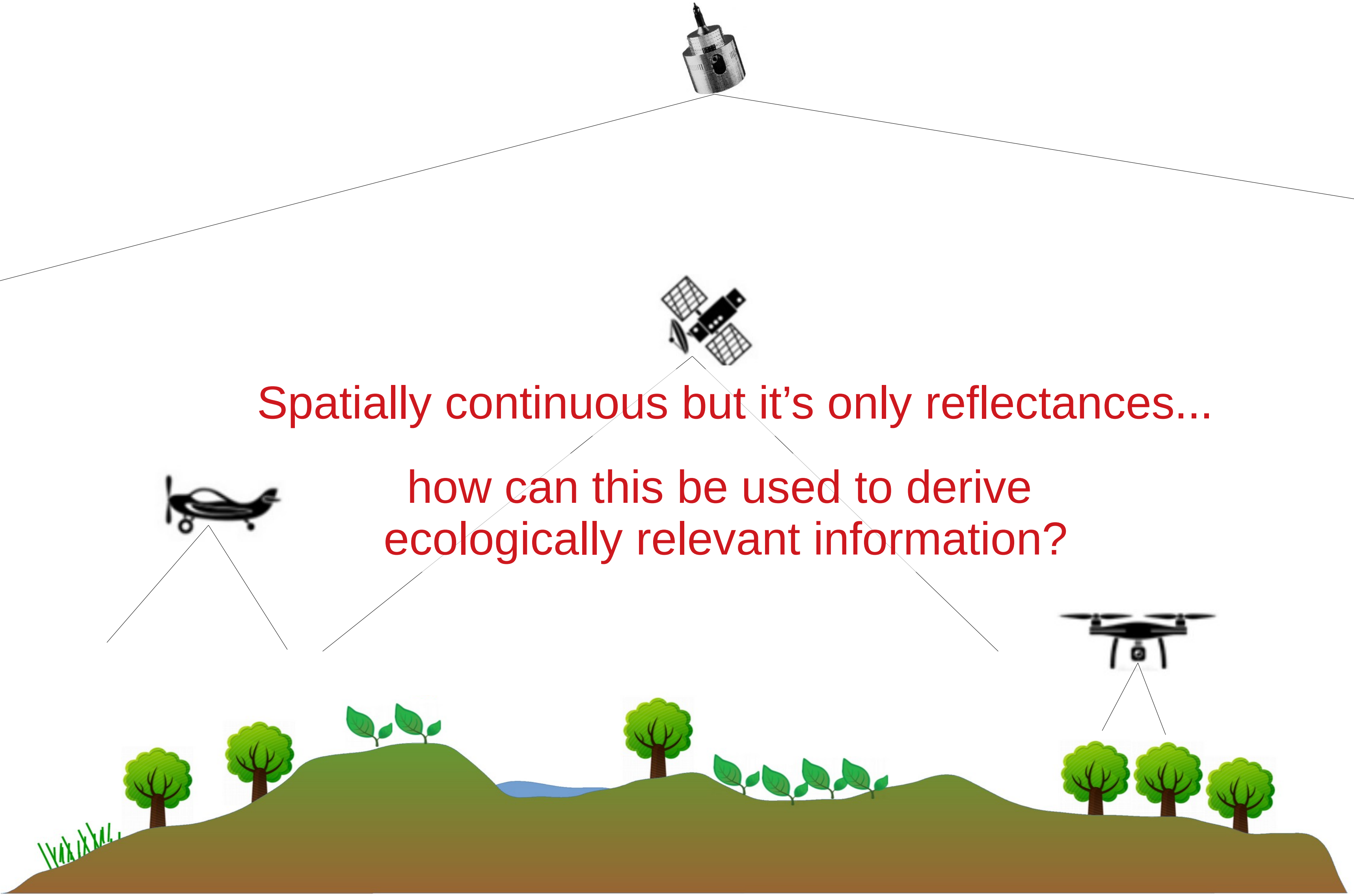




Problem: From field observations to maps of ecosystem variables

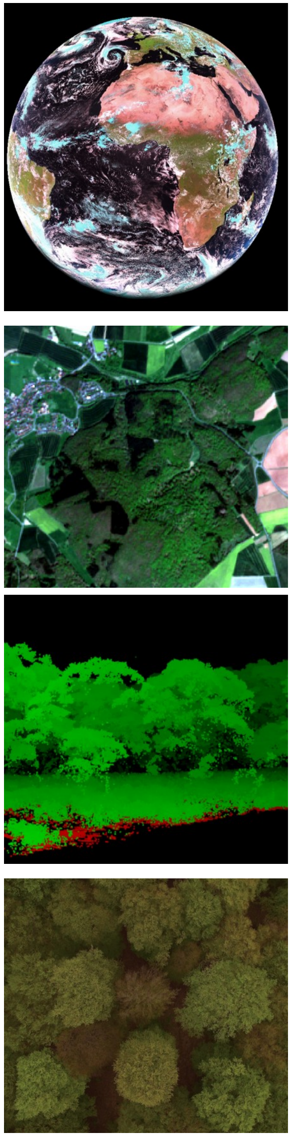


Remote Sensing to derive continuous information



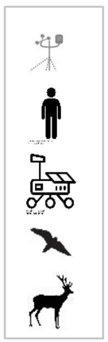
Spatially continuous but it’s only reflectances...

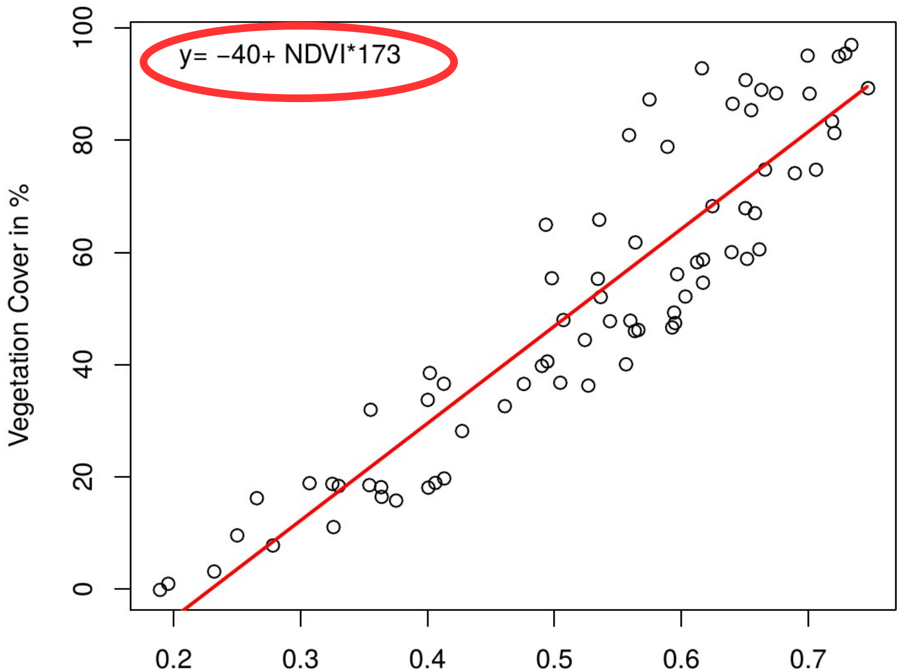
how can this be used to derive
ecologically relevant information?

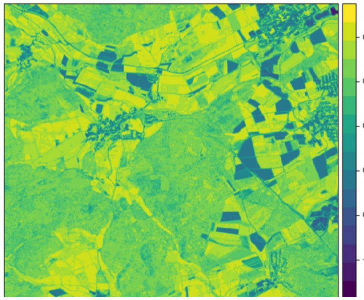


Statistical modelling

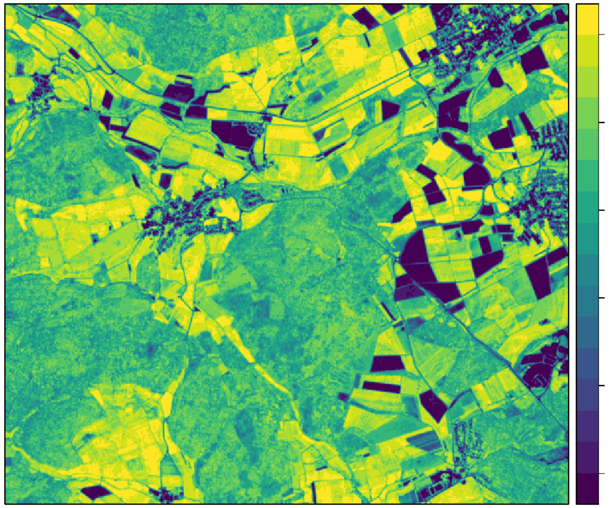
e.g. vegetation cover from remote sensing





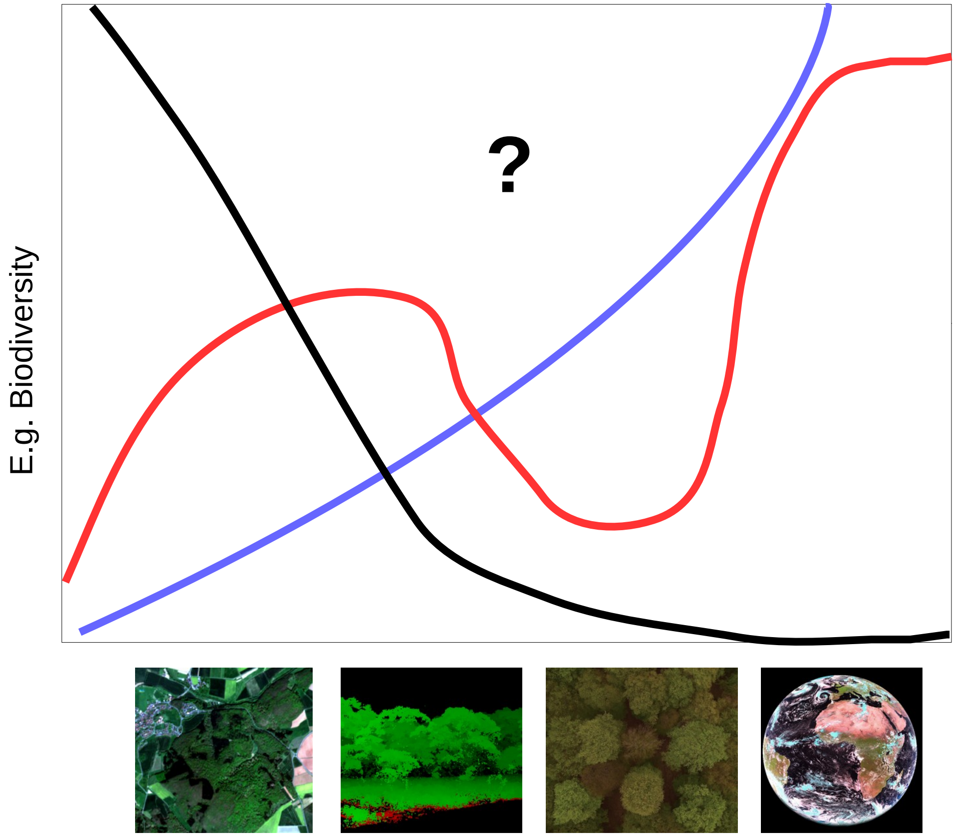


Predicted Vegetation Cover



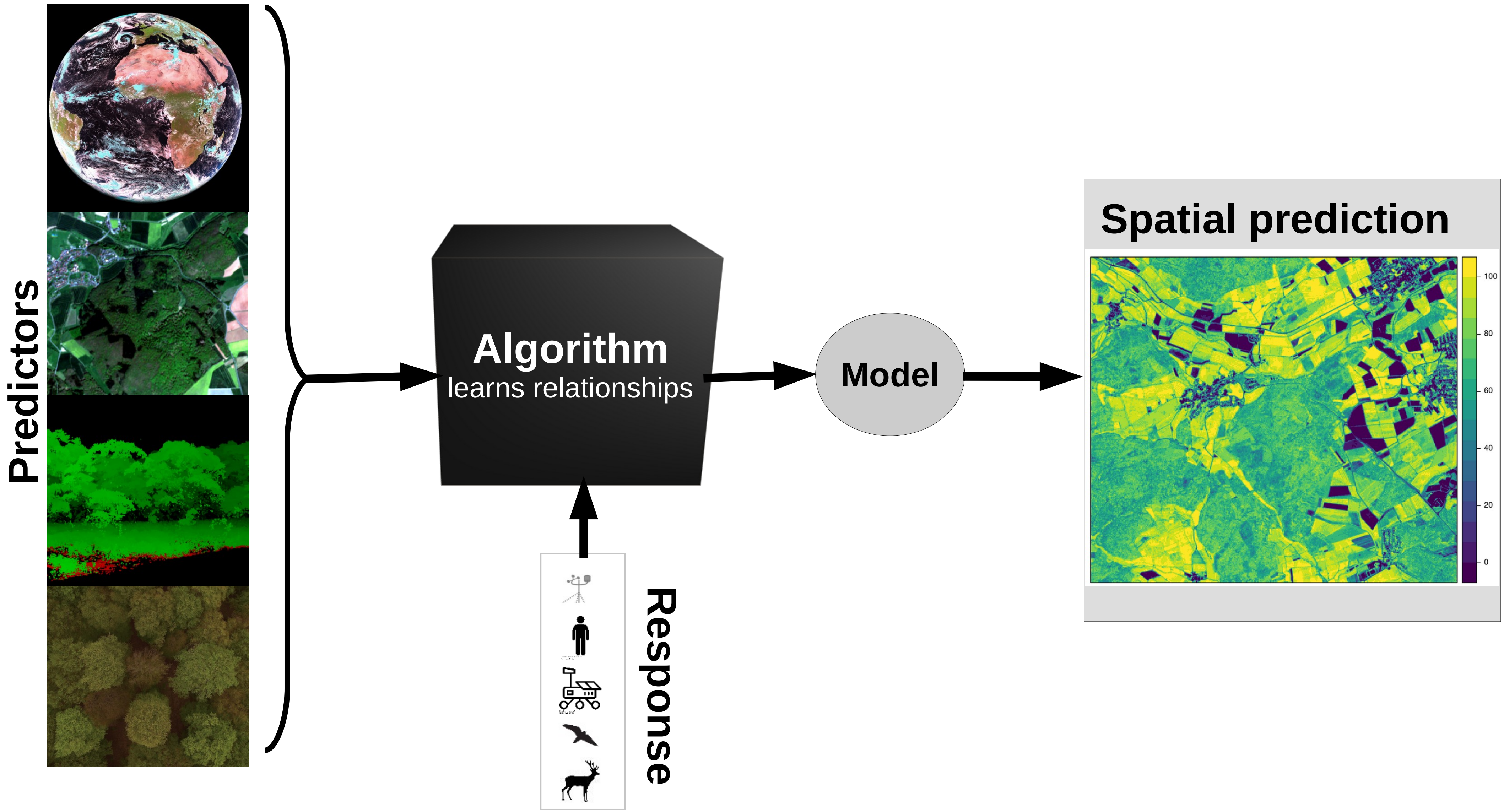
Statistical modelling

… but what about more complex envirnmental variables?



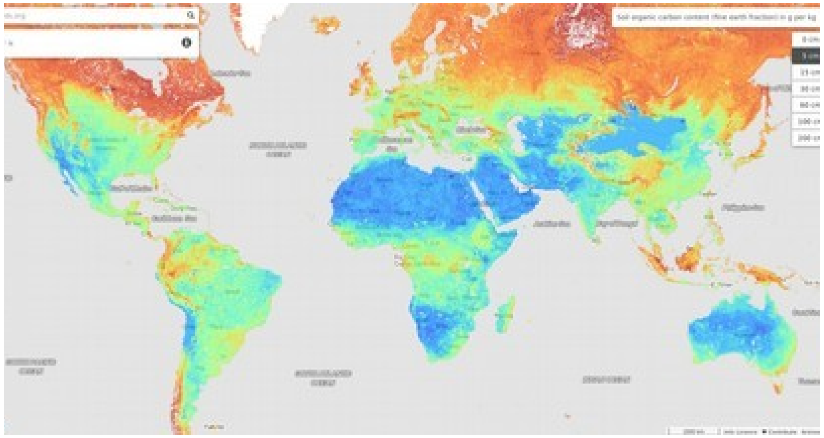
We need models that are able to deal with complex nonlinear relations

From field observations to maps of ecosystem variables

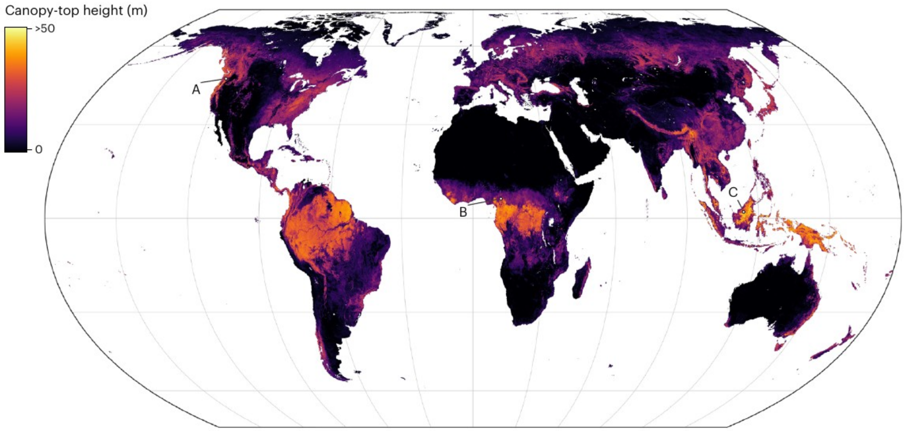


Global maps of ecosystem variables based on machine learning (a few examples)

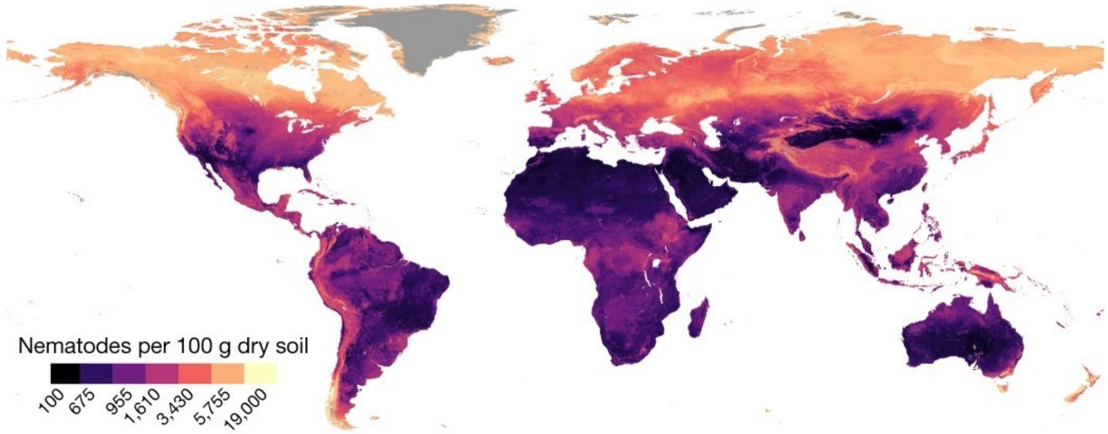
Soil organic carbon (Hengl et al., 2018)



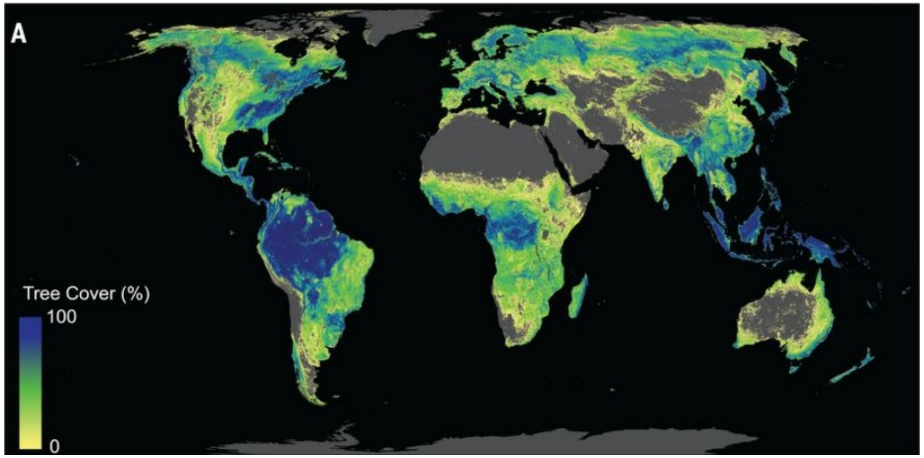
Canopy height (Lang et al., 2023)



Abundances of Nematodes (van den Hoogen et al., 2019)

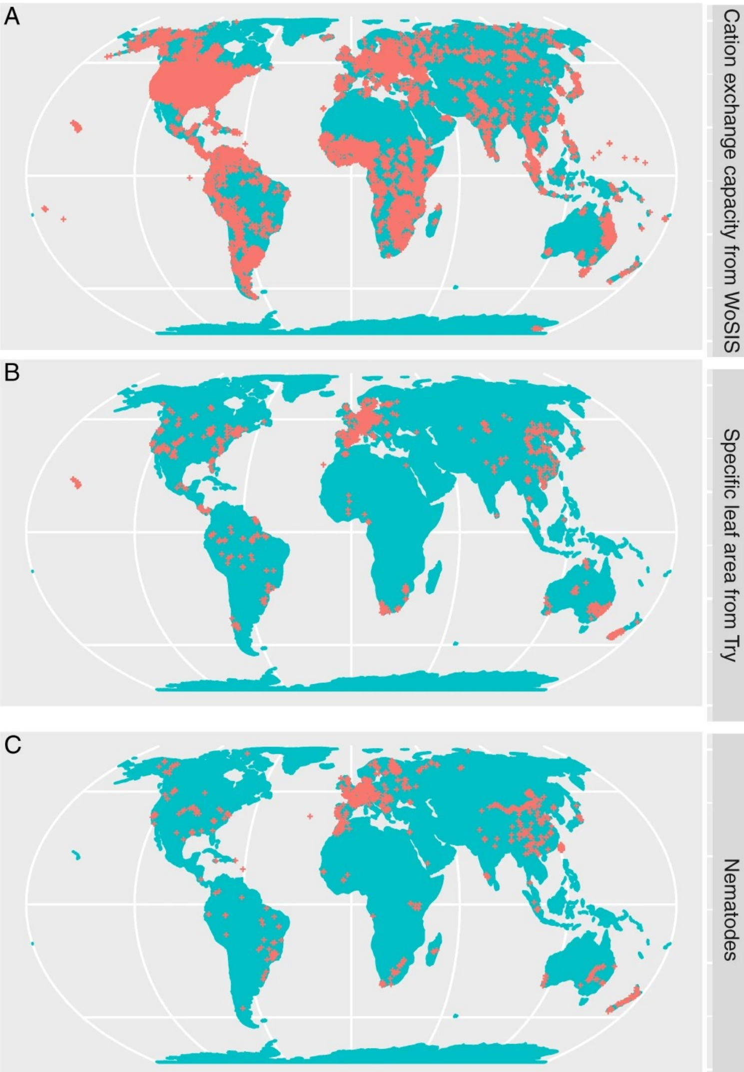


Potential forest cover (Bastin et al. 2019)



Machine learning as a tool to map everything ?

But the models are trained on very few and biased samples



Meyer & Pebesma (2022)

Researchers Find Flaws in High-Profile Study on Trees and Climate

Four independent groups say the work overestimates the carbon-absorbing benefits of global forest restoration, but the authors insist their original estimates are accurate.

Oct 17, 2019
KATARINA ZIMMER

Comment | Published: 23 August 2021

Conservation needs to break free from global priority mapping

Carina Wyborn & Megan C. Evans

Nature Ecology & Evolution (2021) | Cite this article

Wissenschaft

Wenn die KI daneben liegt

Welche Fehler drohen, wenn Forscher Wissenslücken per Computer schließen wollen, zeigen zwei aktuelle Klimastudien.

Von **Tin Fischer**

6. November 2019, 16:44 Uhr / Editiert am 9. November 2019, 17:42 Uhr / DIE ZEIT Nr. 46/2019, 7. November 2019 / 9 Kommentare

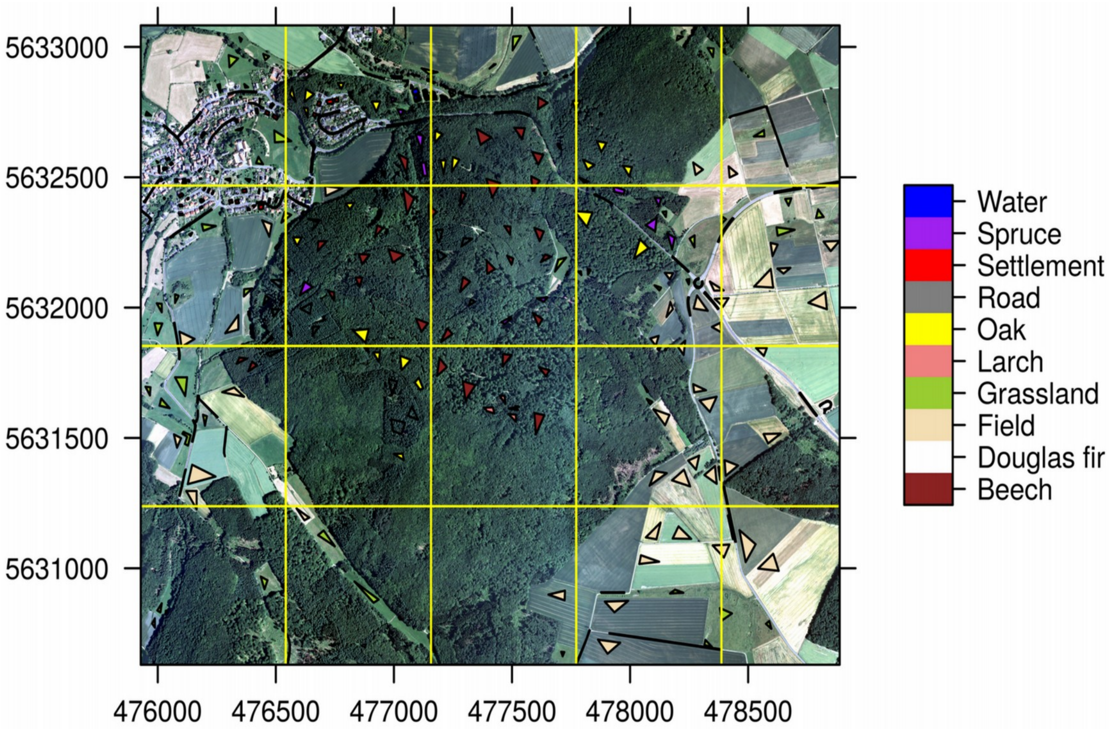
DEEP TROUBLE FOR DEEP LEARNING

BY DOUGLAS HEAVEN

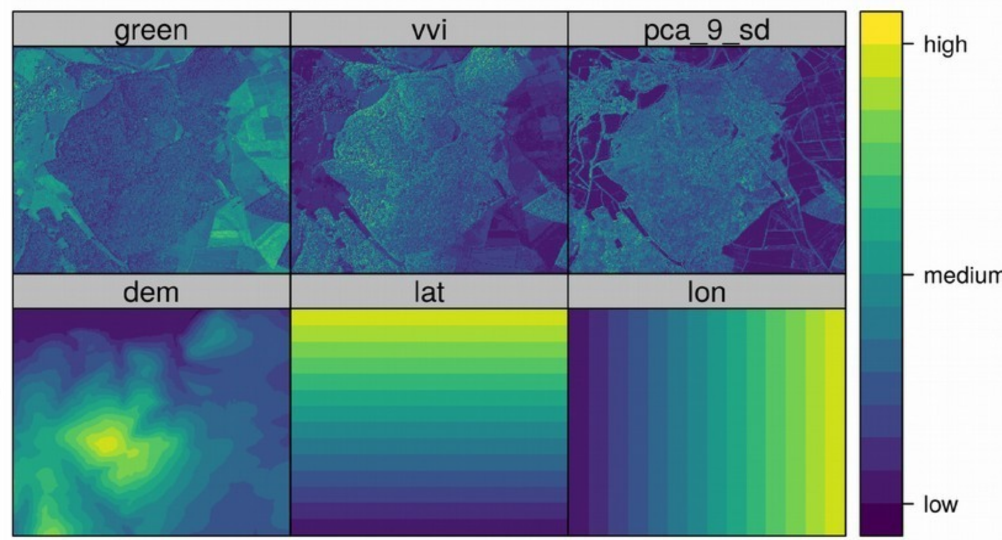
Have we been too ambitious?
When and why might the models fail?

Example of a land cover classification

Aerial image overlayed by training sites



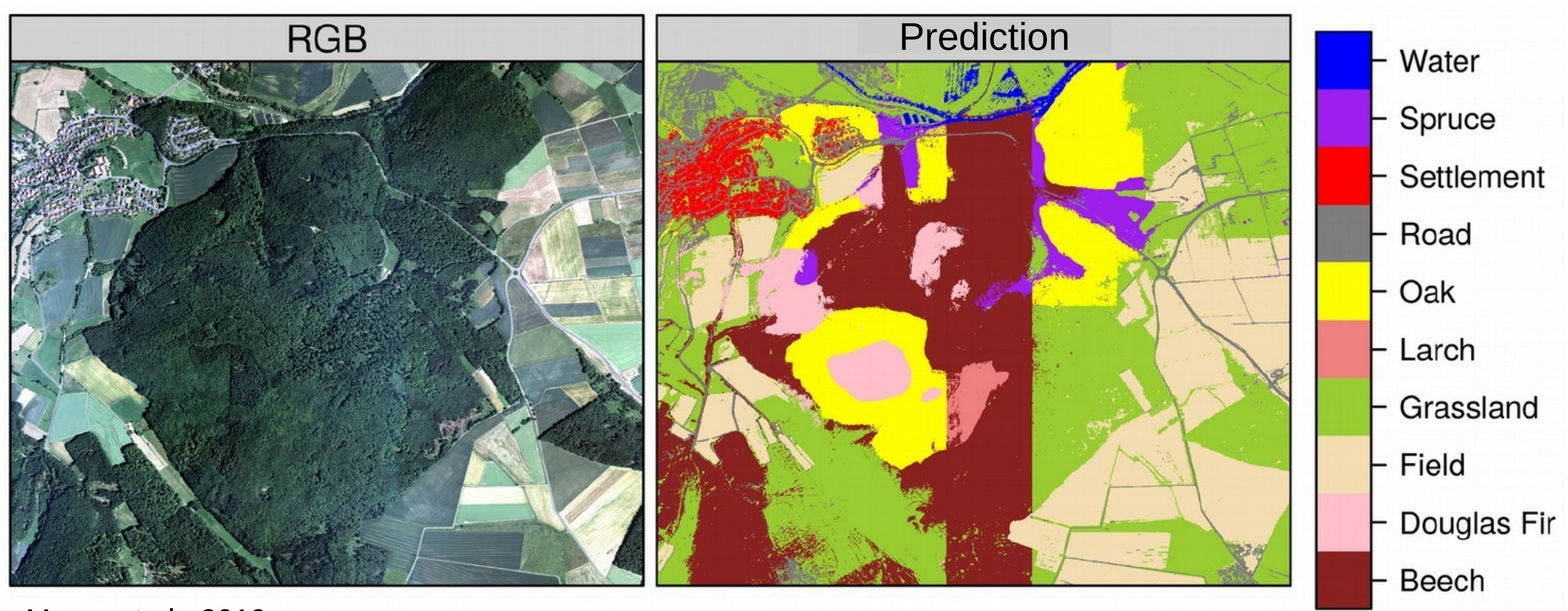
Example of predictors



Random Forests

How well can we model land cover with this approach?

That doesn’t look like a perfect prediction



Meyer et al., 2019

Validation	Accuracy	Kappa
random	>0.99	>0.99
spatial	**0.68**	**0.61**

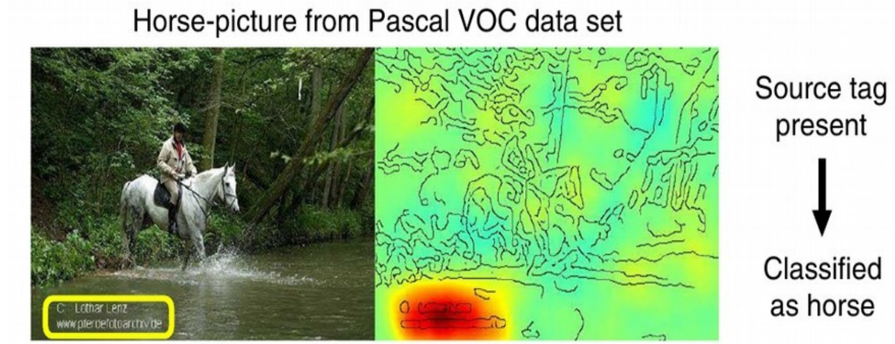
Prediction domain adaptive validation (Linnenbrink et al., 2024)

Why such a poor performance and how can we improve it?

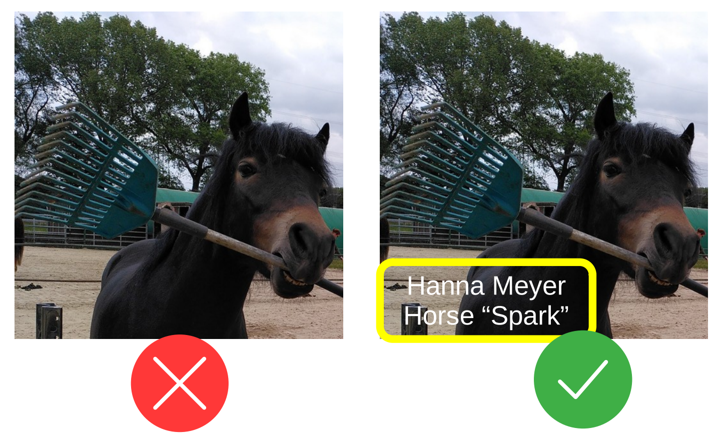
Problem: The model cannot well generalize...but why?

Is the model behaving like the "clever Hans" ?

https://commons.wikimedia.org/wiki/File:Osten_und_Hans.jpg#/media/Datei:Osten_und_Hans.jpg



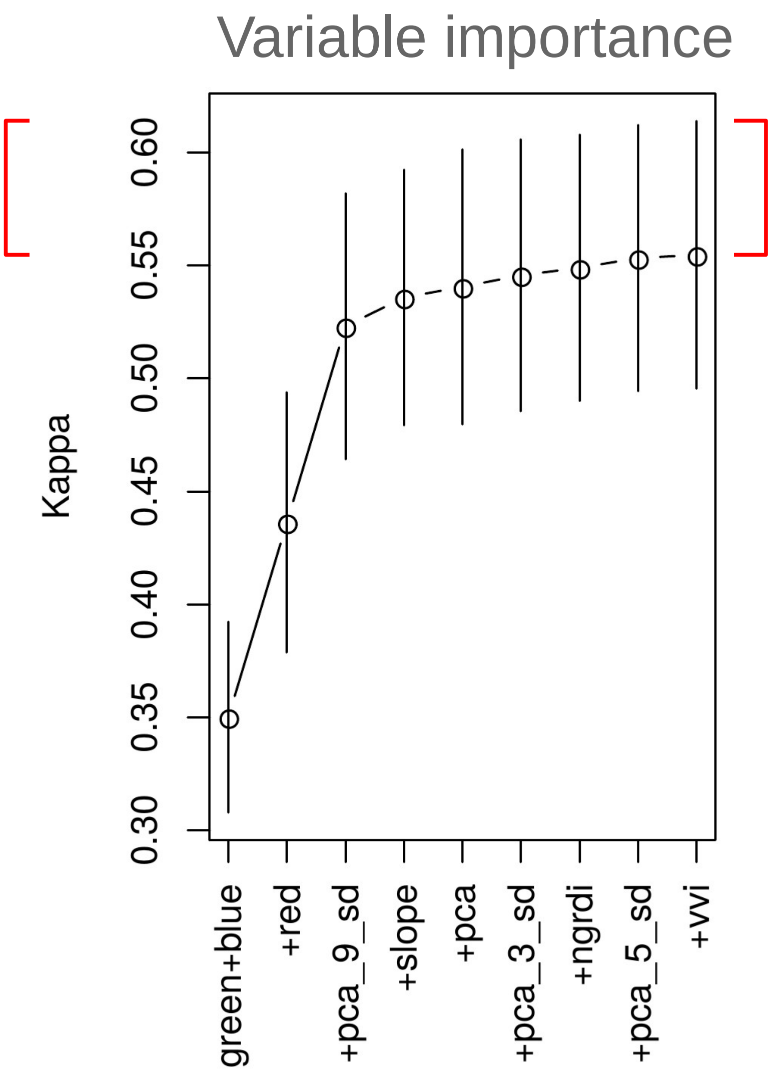
"Unmasking Clever Hans predictors and assessing what machines really learn" (Lapuschkin et al., 2019, Nature communications)



Horse?

Models that are not able to learn the scientifically meaningful relationships → not transferable!

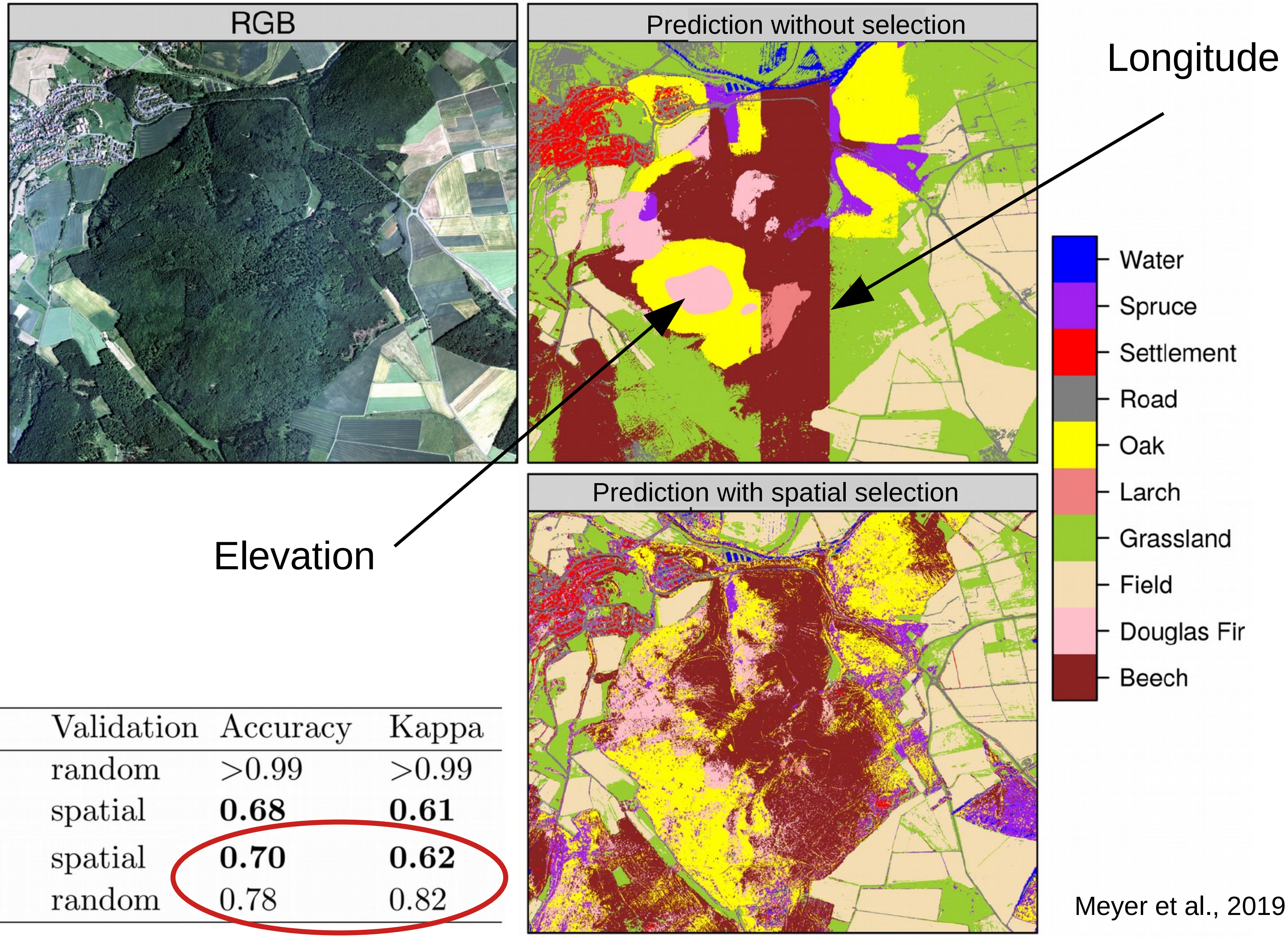
Unmasking “clever Hans predictors” to improve the model?



- Assumption: some of the variables are “clever Hans predictors”
- Removing those variables should improve the results
- Spatial variable selection required!

Implemented in our R package “CAST”

Unmasking “clever Hans predictors” to improve the model?

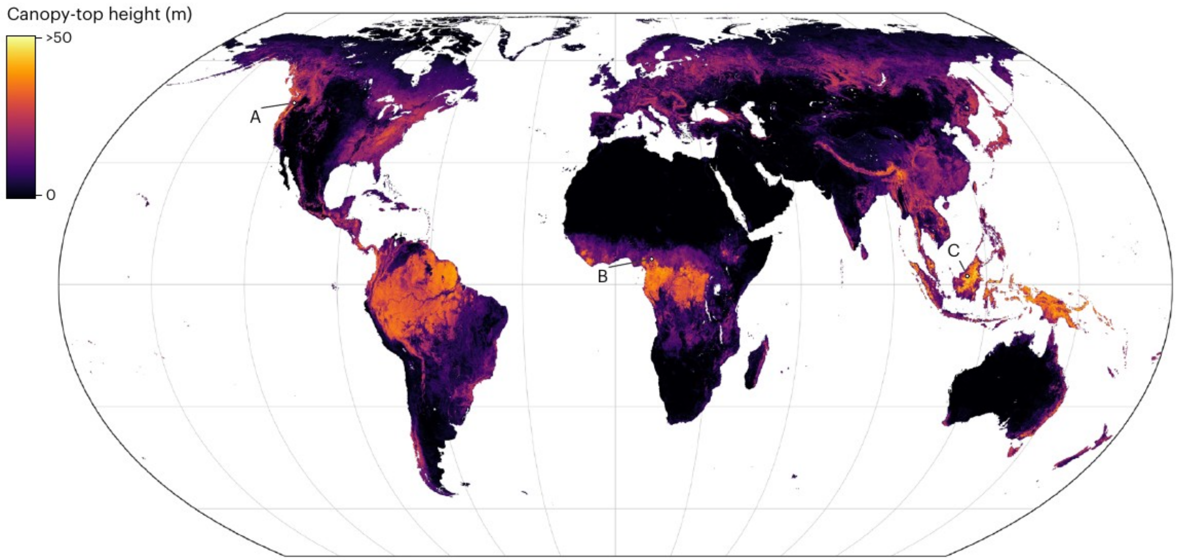


Variables	Validation	Accuracy	Kappa
all	random	>0.99	>0.99
all	spatial	**0.68**	**0.61**
selected by FFS spatial	spatial	**0.70**	**0.62**
selected by FFS spatial	random	0.78	0.82

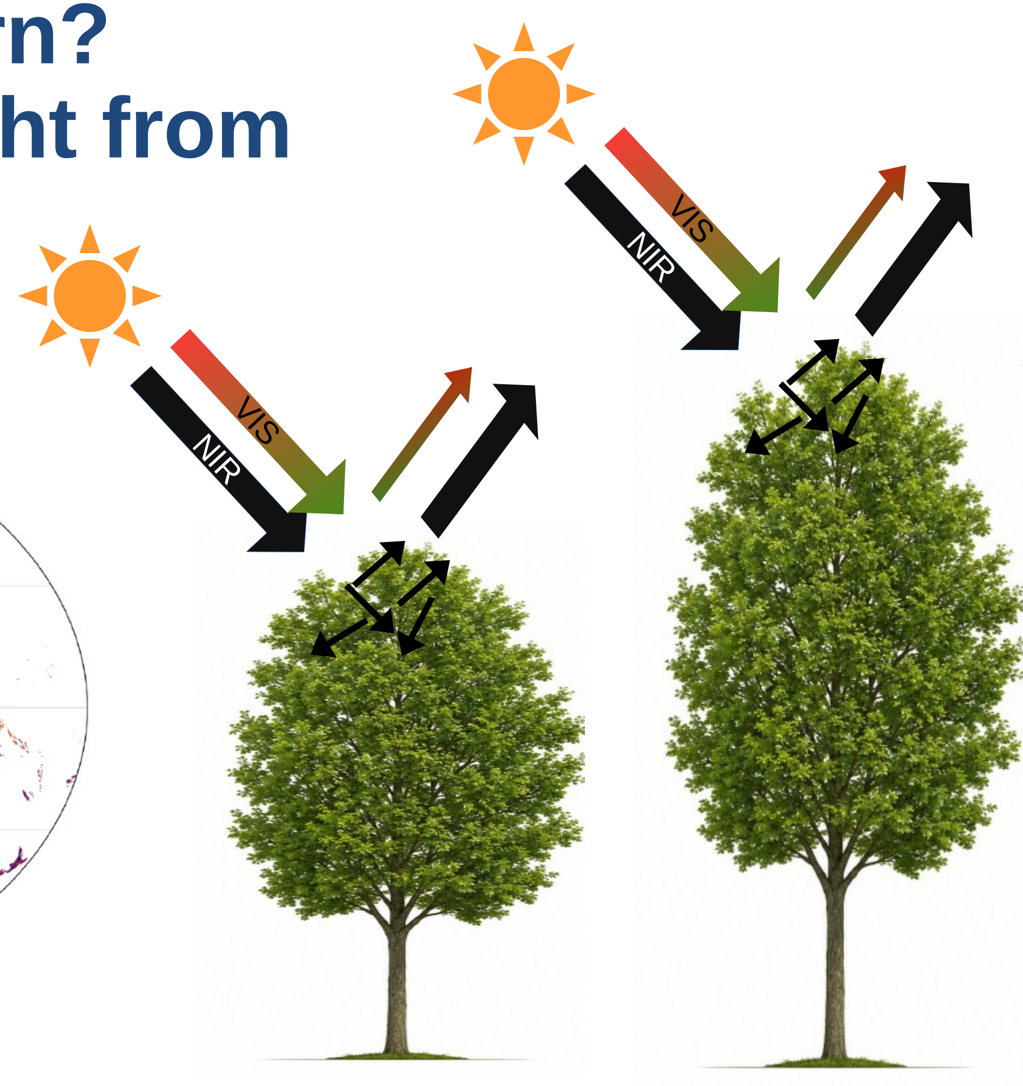
Meyer et al., 2019

What do the models learn? Example of canopy height from optical data

Prediction based on Sentinel-2 satellite data

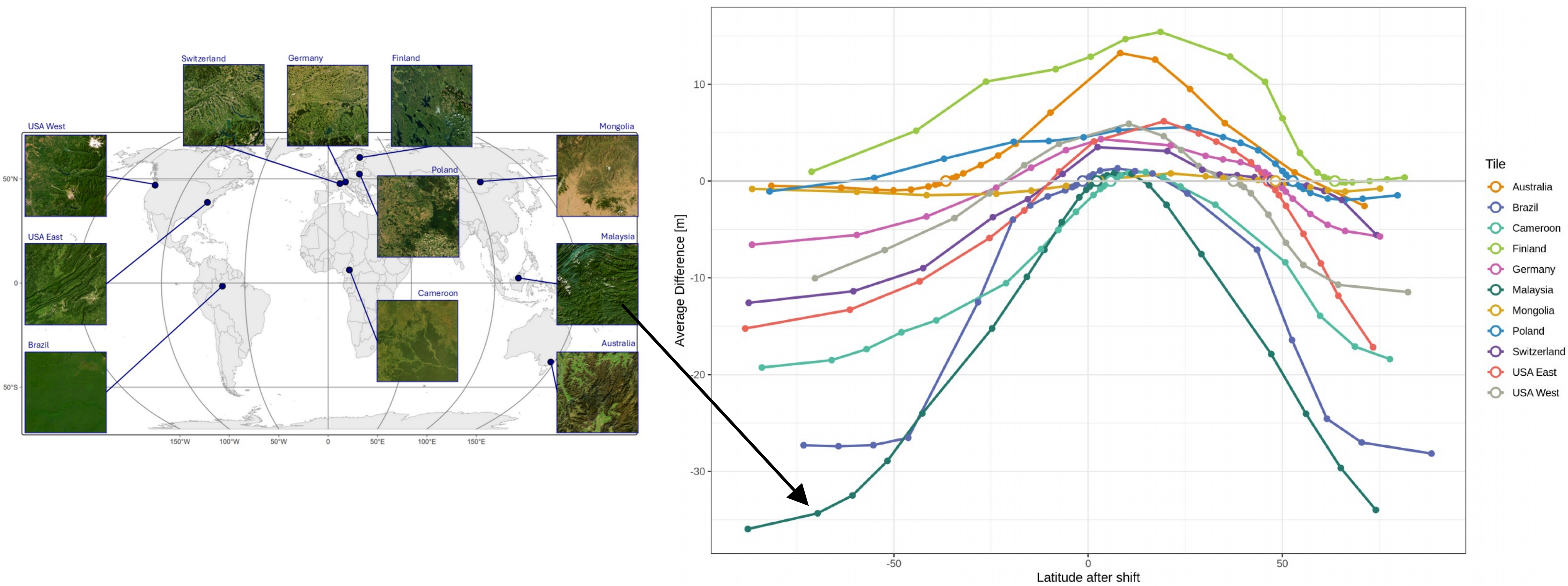


Lang et al. (2023)



Reflectance in the optical wavelengths is independent of canopy height...So what has the model learned?

What do the models learn? Example of canopy height from optical data



Sanchez, Zerres et al (in prep)

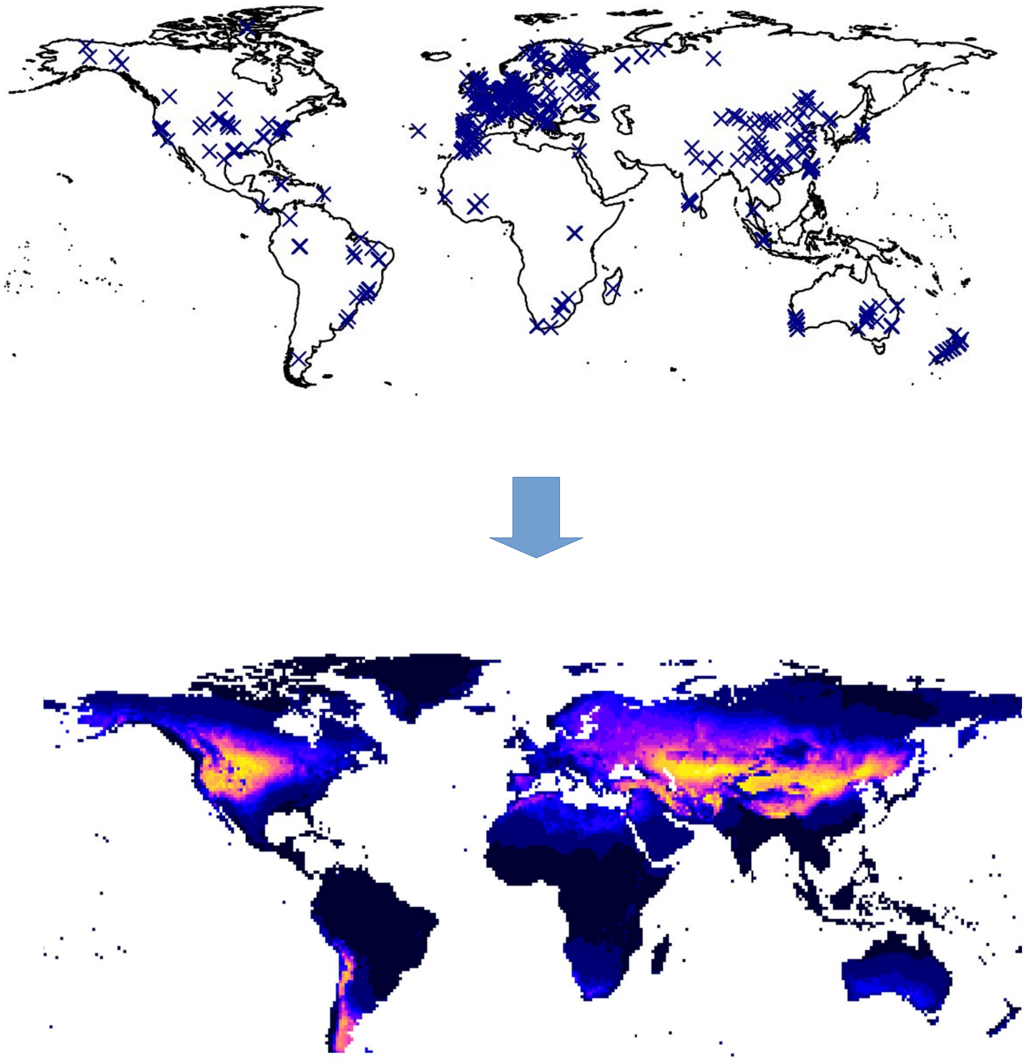
What we have learned so far

- Models often learn shortcuts (clever Hans effect) rather than scientifically meaningful relationships
  → poor transferability

- Variable selection and domain knowledge help reduce shortcut learning

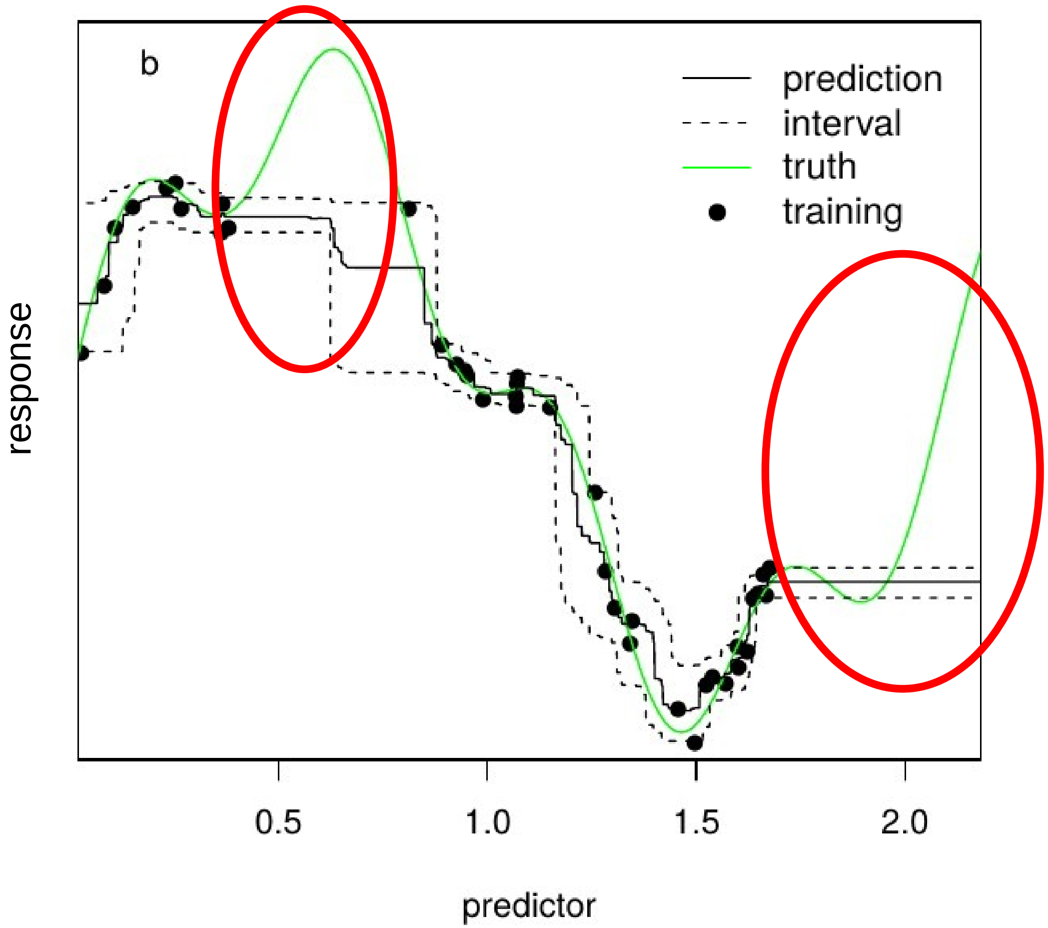
- But even if a model captures meaningful relationships, does that guarantee transferability to a region of interest ?

Is this sufficient for reliable mapping ?



- Transfer to new space required
- New space might differ in environmental properties
- But what if the algorithm has never “seen” such properties?

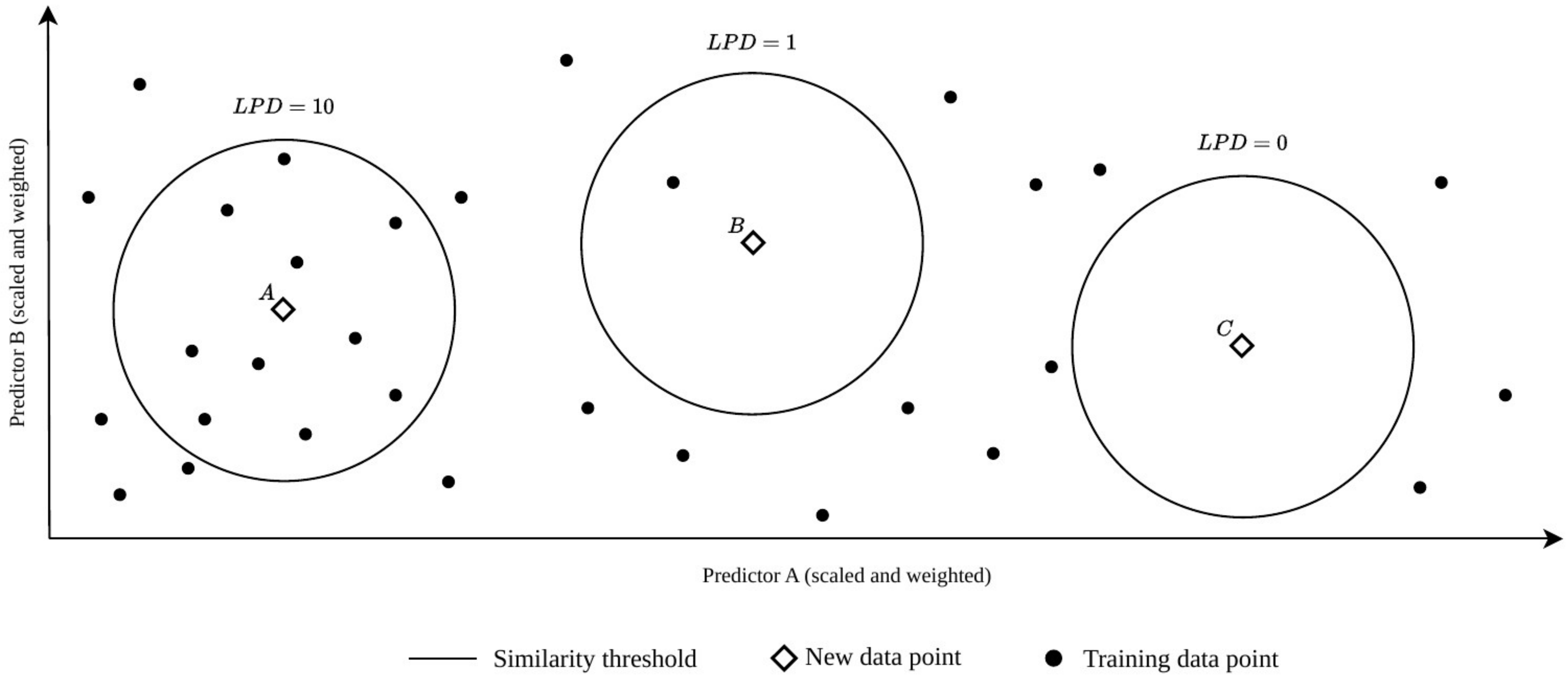
Machine learning models are weak in extrapolations



- Machine learning can fit very complex relationships.
- But gaps in predictor space are problematic (the model has no knowledge about these areas!)
- A measure for “unknown” is needed!

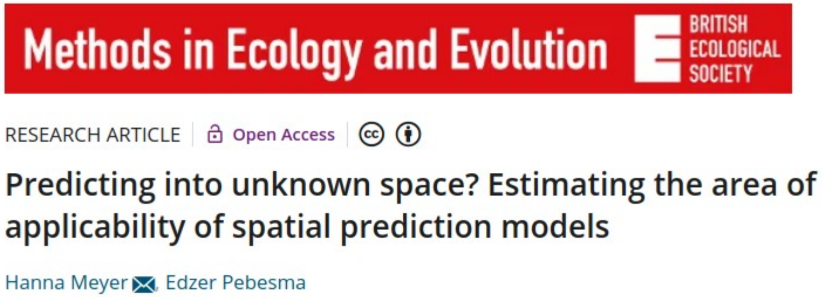
Meyer & Pebesma (2021)

Suggestion of local training point densities (LPD)



Schumacher et al., 2024

Suggestion: Area of Applicability (AOA)


Methods in Ecology and Evolution | BRITISH ECOLOGICAL SOCIETY

RESEARCH ARTICLE | Open Access

Predicting into unknown space? Estimating the area of applicability of spatial prediction models

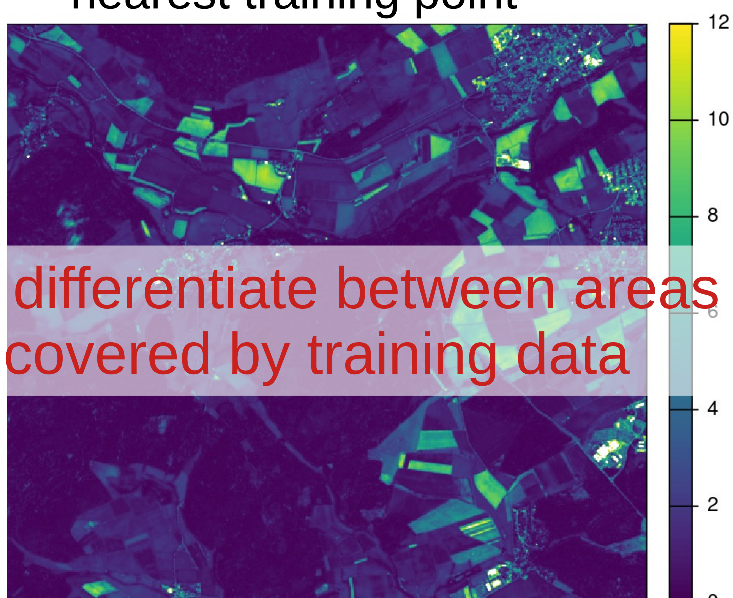
Hanna Meyer, Edzer Pebesma

Based on distances in the predictor space, we try to derive the area...

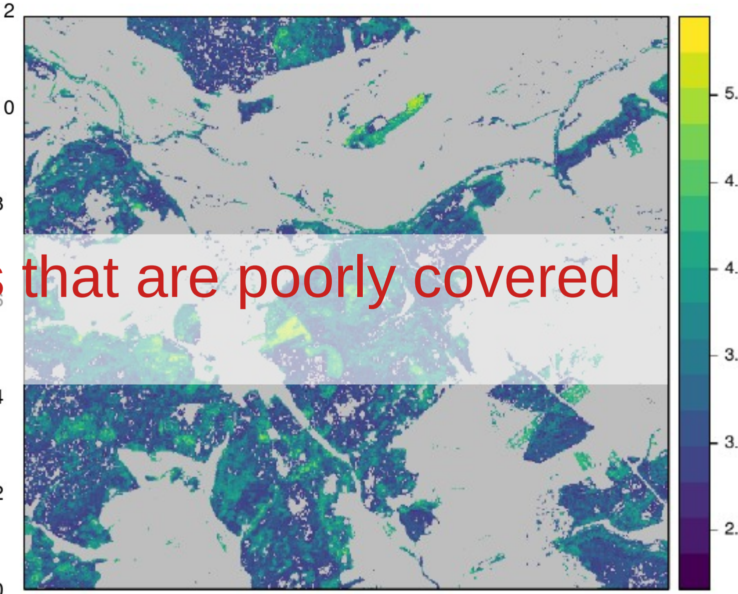
- to which the model can be applied because it has been enabled to learn about relationships
- where the estimated performance holds

Sentinel-2 scene and training data points of leaf area index

Dissimilarity Index based on distance to the nearest training point



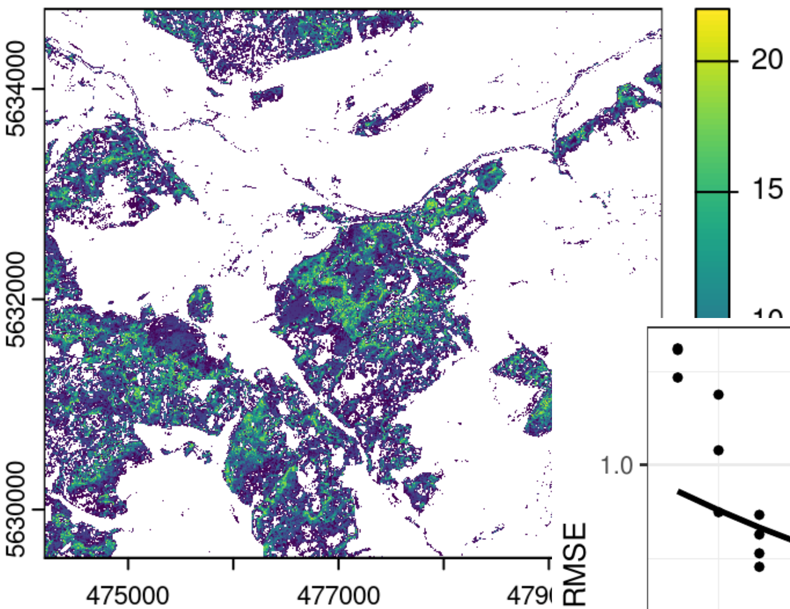
Predictions limited to the AOA



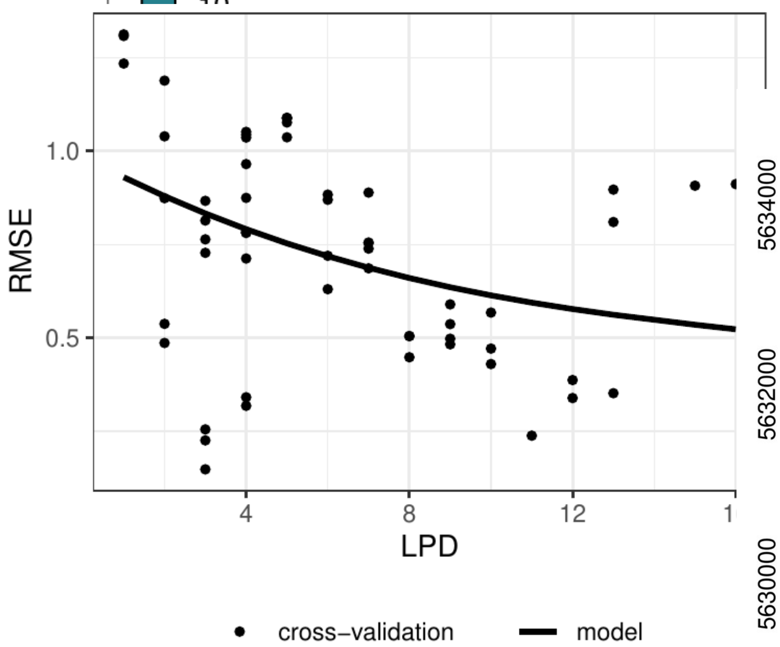
...but the AOA does not differentiate between areas that are poorly covered and areas that are well covered by training data

Densities in the predictor space as a measure for uncertainty?

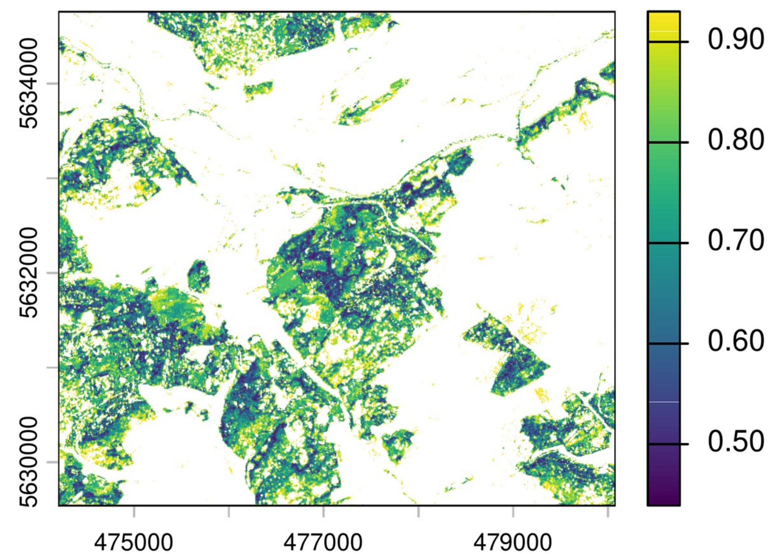
Local Point Density within AOA
(Schumacher et al., 2024)



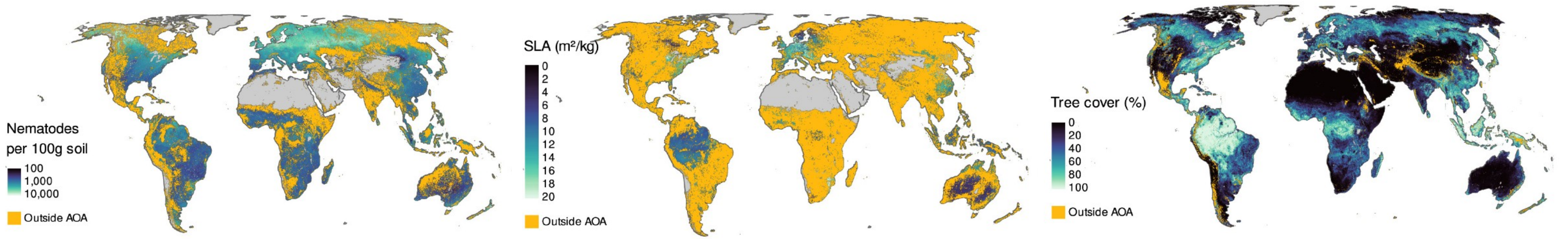
RMSE~LPD



Estimated RMSE



Conclusion



Ludwig et al., 2023

- Results are not just maps; they support modeling, conservation, risk assessment, and more

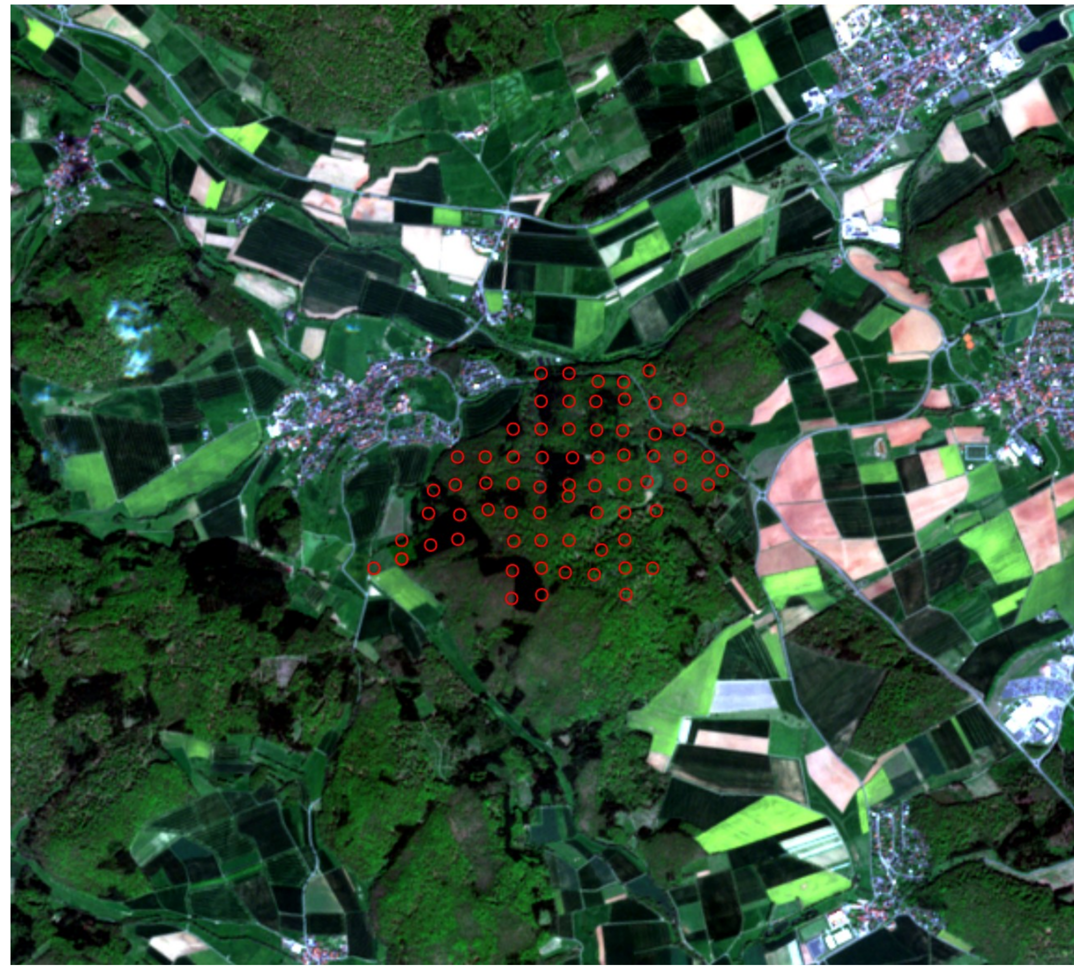
- We, as map producers, must clearly communicate appropriate usage. Don’t leave interpretation to the user

- Identifying gaps isn't a flaw. It’s an opportunity to refine and improve future models towards more comprehensive monitoring of the environment

Time for practice

Response: LAI measurements

Location of reference data



How can we derive spatial continuous LAI data?

https://earth-observation-network.github.io/EON2026/block4_machineLearning/ (part 2)

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